Supplementary material

A regenerative hydrogen cycle for post-release buoyancy compensation in heavy-lift airships: real-time condensation limits and hold-down operation

Credibility assessment with NASA-STD-7009B and details of the results

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Sections, tables, figures and equations numbered without "S" are those of the article.

S1 Purpose and framework

This supplement assesses the credibility of the model of the article and of its results with the NASA standard for models and simulations, NASA-STD-7009B [1]. The standard is written for models that support NASA decisions; any other model may use it as a guide (its Appendix D.1), as here. The assessment follows its structure: the intended use and criticality (this section), the assumptions of the model (Section S2), the phenomena that drive its results (Section S3), the capability of the model, built in its development (Section S4), the results of its use (Section S5), and the outcome, with the warnings for the users of the results (Section S6). Sections S7–S9 then give the details of the results that the article only summarizes: the ideal bound and the altitude control, the mass budget and the heat-rejection classes, the hold-down device and the deliveries in releases, and the energy of the release and of the regeneration.

Intended use The model sizes the requirements that the black boxes of the cycle must meet—burn rate, heat to reject and its rate, response time, mass per kilowatt, hull air flow, and hydrogen and electricity per tonne released—for payload releases in hover from the four airships of Table 2, with no device chosen. Its results inform a decision at the conceptual stage: whether the cycle is worth developing for buoyancy control, and which black box sets the development effort. The model is not meant for the design of the altitude controller, for flight clearance or for the safety of the installation.

Criticality The criticality matrix of the standard (its Appendix D) crosses the consequence of a wrong decision with the influence of the model on it. Here the consequence falls on the cost and schedule of a development, minor to moderate (II–III), with no flight or safety decision. The influence is significant (4): no data exist for the real system, but some exist for similar ones—the water recovery of a hydrogen–air fuel cell [2], the flight tests of a small blimp [3] and the water recovery of the rigid airships [4]. Both cells of the matrix lie in its yellow band, where adherence to the standard is at the discretion of the user.

Assessment The standard assesses separately five capability factors, built in the development of the model, and six results factors, of its use, each at a level from 0, insufficient evidence, to 4. A level is a step without partial credit, reached only when all its conditions are met; the factors are not combined, and the standard sets no level to reach. Level 2, at which a factor rests on formal,

documented evidence rather than informal, is taken here as the threshold for a conceptual decision in the yellow band of the matrix: it asks for formal evidence, but not for the independent review and the validation on the real system that a flight decision would need. The levels are assessed by the authors themselves and remain to be confirmed by an independent review. The acceptance criteria are, for verification, numerical and closed-form errors below the resolution of the reported figures and, for validation, measured water not above the condensation bound, which is an upper bound, and a potential-flow inertia within the measured range. The individual comparisons follow the building-block approach of the AIAA and ASME guides [5, 6], discipline-specific practices of the kind to which the standard refers.

S2 Assumptions and abstractions

Table S1 records the assumptions and abstractions of the model with their rationale and their consequence for the intended use. Those marked favorable make the cycle look better than it would be, so the requirements derived under them are lower bounds of what real equipment must meet; those on the controller are not conservative and leave no margin in the installed rate.

S3 Phenomena identification and ranking

The phenomena that drive the reported quantities were identified and ranked by importance (Table S2), and each was validated against measurements, checked against reference data, bounded with data, shown by sensitivity to be unimportant, or declared outside the scope of this sizing study. Every source was traced to its primary document and graded for reliability (Section S4.2). The word *validated* is kept for comparisons of model outputs with measurements; the inputs are checked against reference data.

S4 M&S capability

The five capability factors assess the model as it was developed: verification, data pedigree, validation, technical review, and process and product management.

S4.1 Verification

The comparison with an exact solution, the numerical uncertainty of the reported quantities and the closed-form depth reproduced by the full model are in Section 8 of the article, with the split of the release with a hold-down device against full simulations (Section 5). With conservation of mass and energy in every simulation, the closed forms against simulation, the consistency between scripts that compute the same quantity and each step of the derivations against an independent computation, they run as automated checks of the pipeline that regenerates every number, table and figure of the article and of this supplement. Of the 1122 checks, 129 compare a value with its reference within a stated tolerance, 330 check a condition (an inequality, a sign or a bound) and 663 require two computations of the same quantity to agree to rounding; these last guard the consistency of the pipeline rather than its accuracy. The export of the numbers stops if any check fails, and all meet their criteria. Verification is level 3: rigorous practices evaluate the end-to-end model and all the important errors meet the criteria. Level 4 would require reliable error estimation for the whole model, such as an independent second implementation.

Table S1: Assumptions and abstractions of the model, their rationale and their consequence for the intended use (Section S1). Favorable: the cycle looks better than it would be, and the requirement derived is a lower bound. Sections, tables and figures without "S" are those of the article.

Assumption or abstraction	Rationale	Consequence for the intended use	Treated in
Vertical motion only; payload released as a step change in load, from rest at equilibrium	The sizing maneuver is a release in hover; a step change is the fastest release	Conservative: a gradual release needs a lower burn rate and heat-rejection power, and a set-down by winch over a time T follows Equation (8) with $t_{\mathrm{rec}} = T$; attitude coupling is not represented	Section 4
Two regimes: buoyancy control alone, then with a hold-down device	Design premise of the cycle: the burn is the only means that removes lift. The first regime bounds what the cycle must deliver by itself; in the second, a device holds the vehicle while the burn proceeds	Conservative in the first regime: any means acting on the vehicle relaxes the requirement. The second shows how much: the power becomes a choice of recovery time, and the burn absorbs by itself only the part m_e of the release	Sections 1, 4 and 5
Hold-down device: ideal propellers with a figure of merit, powered by the cycle's engine or, beyond its reach, by the plant of the airship; response bounded for a release from rest or for a force ramped from before it	No device is chosen; vectored propellers are the means most airships carry, and a tether the means of a moored airship	Favorable for the propeller power and energy, since heavily loaded real propellers have lower figures of merit; conservative for the response, since a burn before the release gives the device more time. The response of a real device and the lightness it can be certified for are not validated	Section 5; Section S8
Storage tanks as an ideal source at up to $\dot{m}^*_{H_2}$; regeneration outside the maneuver	Regeneration takes 24–26 h per tonne released with electricity from outside; the tanks only return hydrogen in the trim	Favorable for the trim, since a real regulator lags; no effect on the sized burn, which returns no hydrogen. The regeneration and storage boxes are sized by energy and mass balances in Sections 5.1 and 6, not simulated	Sections 2, 5.1 and 6; Appendix B
Equipment masses nominal, those of the lightest classes found and of two representative classes of panels, with no growth margin	Requirements analysis at the conceptual stage: each black box is represented by a class of existing equipment, not by a design	Favorable: a growth margin on the equipment would leave less of the payload for the release and the delivered cargo; 10 % on all of it costs 2.4 t of the 36 t of the reference delivery	Sections 5.1, 6 and 8; Table 5
Ideal bound: instantaneous condensation with all the water retained ($\phi = 1$), heat of condensing $\phi = 0.99$	Separates the burn-rate requirement from the condensation requirement	Favorable: a real unit retains less and needs a higher rate; at 1 atm at most 0.85 of the water condenses	Section 5; Section S5.3
Heat-rejection unit: first-order lag that cools completely the fraction of the exhaust it can handle	No device is chosen; no response time is published for a unit of this kind	Favorable: a unit in series condenses less; τ is bounded with estimates and one measurement	Sections 3.4, 8
Stoichiometric combustion ($\lambda = 1$)	Most concentrated vapor, highest dew point	Favorable: real combustors often operate with excess air, which dilutes the vapor and retains less water	Figure S1; Section 9
Ideal, isothermal lifting gas at the temperature of the air	Thermal time constant of about 27 min against a burn of seconds; an ideal gas changes the hydrogen per tonne by 0.04 %	Negligible for the sized burn; not valid over hold-down recoveries and deliveries in consecutive releases, of minutes to hours, where the thermal drift is of the order of a release or of several	Sections 9, S6
Lift following the air density with the gas volume fixed	Simplicity of the simulation	Adds 61 % to the loop stiffness; with the gas cells about half full the real lift barely changes with altitude, so the trim is faster than in the airship; the excursions, set by the saturated burn, are unaffected	Section 3.2
Release at the planned instant, mass on board known to the controller; gusts neglected	Isolates the sizing of the burn from estimation and disturbance rejection	Not conservative: the installed rate carries no margin for timing errors, mass estimation or gusts	Section 8; controller design
Hull admits air without delay; conversion unit running when the maneuver starts	The flow and the start are stated as requirements of other black boxes	Requirements of the hull air system and of the start sequence, not limits of the cycle	Sections 2, 5.1
Standard atmosphere at sea level; 30 °C design day for the thermal balance	Buoyancy depends only on the density ratio of gas and air	The reference density changes the drag and the inertia by 0.5 %; the 30 °C day would lower the inertia by 5 % and raise the installed rate by 2.6 %; hotter days retain less water	Table 4; Section S4.2
Dry intake air	Simplicity; the humidity changes with the day	Conservative: at 50–90 % relative humidity the intake brings 0.46–0.84 kg of water per kg of hydrogen, which condenses and lowers the hydrogen per tonne by 1.6–2.9 %	Section 3.5
Unpublished airship data: the LCA60T taken with its published	Only published or official data are used; a missing datum is	Conservative for the LCA60T: the spheroid of its published dimensions encloses 1.45 times more, which would lower its dynamic requirements by about 17 %; the one tonne sets the	Table 2, Section 3.5

Table S2: Phenomena identification and ranking (PIRT) for the reported quantities. Importance: H, changes a reported requirement by more than 10 % or the feasibility conclusion; M, by 1–10 %; L, by less than 1 %; in parentheses, outside the sized maneuver. Treatment: how the phenomenon was addressed in this study.

Phenomenon	Quantity affected	Imp.	Treatment	Evidence and result
Buoyancy–mass balance, thermochemistry (L, q)	H_2 per tonne; P_{rej}	H	Reference data	Thermochemistry reproduced (Table S3); H_2 density within 0.06 %; ideal-gas H_2: 0.04 %; the air density at h_0 is the sea-level reference and cancels in $\rho_{\mathrm{air}}/\rho_{H_2}$
Vertical inertia with added mass (M_I)	$\dot{m}^*_{H_2}$; P_{rej}	H	Bounded with data	Heave mass 0.77–1.20 of potential flow, 18-m^3 blimp [3]: 2.89–3.61 kg/s, 407–509 MW; static results unaffected
Condensation limit $\phi_{\max}$	Water retained; H_2 per tonne	H	Validated	Within 0.12 percentage points of the model of [2]; measured water 68–85 % of Equation (5), an upper bound; 50.3–53.9 kg/t
Outside-air temperature	$\phi_{\max}$; make-up H_2	H	Sensitivity	Table 4; at 1 km, 85 to 89 % of the water retainable
Intake-air humidity	H_2 per tonne; q	M	Sensitivity	50–90 % relative humidity at 30 °C: 0.46–0.84 kg of water per kg of H_2 condenses; H_2 per tonne −1.6–2.9 %, heat per tonne −0.8–1.5 % (dry intake is conservative)
Heat-rejection power and specific power	Release size; mass feasibility	H	Output	Derived (Section 5.1); compared with 7 existing classes at 1 atm (Section S8)
Mass of the other black boxes	kW/kg left for heat rejection	H	Bounded with data; sensitivity	Published ranges (Section S8); at 1 atm, fan fraction $f = 0$ to 0.15: 8.1–93 kW/kg, no budget above 0.16; fans sized for the pressure drop (f = 0.008–0.041): 8–11 kW/kg; best 30-s battery: 12 kW/kg
Authority and response of the hold-down device	Releases held; recovery time	H	Closed form; simulation	Response ceiling of Equation (9); full simulations of a tonne with 10 MW stay within the 1-m threshold and the propeller energy is 1.00–1.02 times the closed form; anticipating the release, the device tolerates 2.3 s of lag on the Zeppelin NT, against 1.2 s waiting; static lightness certified on the Zeppelin NT, 200 kg
Conversion efficiency η	Self-powered cooling; electricity	M	Published range	0.30–0.45 for hydrogen engines [7], above $\eta_{\min}$ = 0.15–0.19
Condenser response time τ	Water retained; start time	M	Bounded; sensitivity	5.8–6.3 s estimated [8], 16–59 s measured [9]; 49–52 kg/t with $\tau = 5$ s and 50–52 kg/t with 30 s; 63 % retained if started with the burn, with unlimited power
Condenser arrangement; stoichiometric air	ϕ; P_{rej}	M	Favorable assumption	Both maximize the water retained (Sections 3.4 and 3.5)
Electrolysis and compression energy	External electricity per tonne	M	Reference data	53 kWh/kg, 2024 average [10]; compression 2.0–3.1 kWh/kg [11, 12]
Lift variation with altitude	Trim	L	Estimate	+61 % loop stiffness; excursions set by the saturated burn unaffected (Section 3.2)
Lifting-gas superheat	Buoyancy drift	L (H)	Estimate	16–65 kg during the sized burn; 0.6–1.2 t over the recovery of one tonne at 10 MW with the exhaust at 14 atm and 3.0–6.0 t over a delivery of 6 h, with a thermal time constant of 31 min for the loaded airship [13, 14]
Drag; atmosphere model	Peak excursion	L	Sensitivity; reference data	C_D = 0–1: 0.5 % at most; sea-level reference state: 0.5 % of the drag and of the inertia
Release timing; mass known to the controller	Peak excursion; clean burn	–	Outside the scope	Release at the planned instant; mass on board known (Section 3.5); left to the controller design
Gusts; hull air admission	Peak excursion; hull flow	–	Not assessed	Gusts neglected; hull flow given as a requirement (Section 5.1)
Numerical integration	Dynamic quantities	L	Code verification	Exact solution within 0.003 %; convergence below 0.002 %; closed-form depth $(1 + s)\, D_{\mathrm{lim}}$ reproduced

S4.2 Data pedigree: reference-data checks

The physical inputs of the model are taken from published reference data, and Table S3 checks each block against its source, a verification of the inputs rather than a validation of the model: the standard atmosphere [15, 16], real-gas properties [14, 17], thermochemistry [18], atomic weights [19], the saturation of water [20, 21] and electrolysis [10]. The higher heating value rebuilt from the lower heating value, the latent heat and the stoichiometry reproduces the tabulated value, and the saturation pressure used for ϕ differs from the reference equation of state by 0.007 % at most. The remaining modeling assumptions change the results little: treating hydrogen as an ideal gas changes the hydrogen per tonne by 0.04 %, and the sea-level reference state changes the drag and the inertia by 0.5 %; the air-density row of Table S3 shows that choice of reference state, not an error, and it cancels in the density ratio of the buoyancy relations.

Every source, of the data of the model and of its inputs, passed a six-step protocol before use—identity with a SHA-256 fingerprint of the document, data located, internal consistency, contrast with a second source or a physical limit for the critical data, findings and decision—and was graded for pedigree, from A, primary reference data that the model reproduces, to C, qualitative or preliminary data, or data without the evidence behind them, such as manufacturer sheets: 10, 28 and 22 of the 60 sources. The register is distributed with the code. Data pedigree is level 2: all data are traced to formal documents with their uncertainty estimated, but the added mass comes from potential flow checked against an 18-m^3 blimp, not a referent representative of the hull.

Table S3: Reference-data checks of the model inputs: model value, published reference and deviation. HHV: higher heating value; ISA: International Standard Atmosphere [15]; IEA: International Energy Agency. Checks that reproduce the source by construction (gravity, oxygen in air and per kilogram of hydrogen) are not listed.

Block	Quantity	Model	Reference	Dev.	Source
Atmosphere	Air density, sea-level reference at h_0 (kg/m^3)	1.2250	1.2191	+0.48 %	ISA
	Density scale height at h_0 (m)	10,416	10,405	+0.11 %	ISA
Gases	H_2 density, 15 °C (kg/m^3)	0.08526	0.08521	+0.06 %	NIST
Combustion	H_2 lower heating value (MJ/kg)	119.96	119.96	+0.00 %	CODATA
	Latent heat of water, 25 °C (kJ/kg)	2441.7	2441.7	+0.00 %	NIST
	HHV rebuilt as LHV $+(M_{H_2O}/M_{H_2})L_v$ (MJ/kg)	141.78	141.79	-0.01 %	CODATA
Saturation	p_{sat} at 40 °C, IF97 (kPa)	7.3844	7.3849	-0.01 %	IAPWS-95
	p_{sat} at 100 °C, IF97 (kPa)	101.418	101.418	-0.00 %	IAPWS-95
Electrolysis	Electricity per kg H_2 (kWh/kg)	52.9	52.9	+0.00 %	IEA

S4.3 Validation

Section 8 of the article runs the condensation bound, Equation (5), against the water recovery of an open-cathode fuel cell [2]: the agreement with their analytical model, the same Dalton balance, is a consistency check, and the water they collected is the experimental test. The details of the comparison are these. With their measured flows, Equation (5) gives 1.5–2.6 % less water than their Appendix A recomputed with the same flows, in experiments 1–5 (Figure S1). Their tabulated expected water differs from that recomputation by 1–12 %, most in experiment 5, and experiment 6 lists a saturation humidity that does not match its temperature, so the recomputation, not the table, is the reference. The water collected, 68–85 % of the bound in their 6 experiments at 25–27 °C, makes ϕ_{max} an upper bound that a real condenser approaches.

The other block compared with measurements is the vertical inertia: the heave mass identified in constrained and free-flight tests of an 18-m^3 blimp [3] is 0.77–1.20 times the potential-flow estimate of Equation (2), a range that contains the estimate (Section 9). Over that range the damping ratio of the altitude loop, ζ divided by the square root of the ratio, is 1.19–1.48, so the loop stays overdamped.

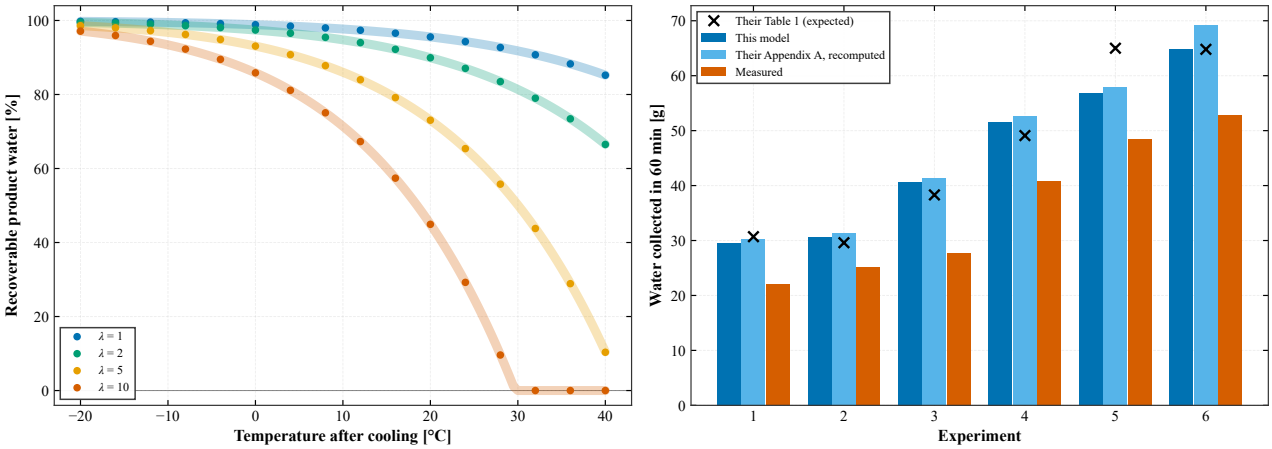


Figure S1: Validation of the condensation bound against Zakhvatkin et al. [2]. Left (consistency check): recoverable product water versus the temperature after cooling, their analytical model recomputed from their equations (thick bands) and Equation (5) of the article (dots), for several excess-air ratios λ; above the dew point of each mixture no water condenses. Right (experimental test): water collected in their six experiments, as predicted by this model with their measured flows, as given by their Appendix A recomputed with the same flows (bars) and as tabulated by the authors (crosses), and as measured.

Validation is level 2: key outputs compare favorably with representative systems within the domain of operation, but no measurement exists for the real system, an airship releasing a payload. Not validated are the integrated vertical dynamics at full scale, the response of a heat-rejection unit of this kind and the controller; they bound the domain of validity (Section S6).

S4.4 Technical review and process management

The model, its data, verification and validation were reviewed informally by co-authors who did not develop it, and its code and sources were audited internally; no formal or independent review has taken place before submission, so the development technical review is level 1. The development is managed as one pipeline under version control: every constant is defined once, every number, table and figure of the article is generated, and the export of the numbers to the manuscript is gated by the verification checks and by a fingerprint of the data they verified. This configuration management uses formal methods, but the process, its roles and the requirements of its products are documented informally and were not formally approved, so process and product management is level 1; level 2 would need that formal documentation and approval.

S5 M&S results

The six results factors assess the runs reported in the article.

S5.1 Use assessment and input pedigree

Every run is within the limits of the model and within its domain of verification, and the proposed use is the intended use of Section S1. It is, however, the first use of the model, which the standard places at level 1 by definition. The inputs of the runs are the airships of Table 2, from the Hindenburg report [22], a NACA memorandum [23], a type certificate [24], a NASA-sponsored historical study [4]

and the manufacturer of the LCA60T [25, 26], and the technology classes of Section S8. Those that size the results come from formal documents, with their uncertainty estimated: the hull volume, with the two figures published for the LCA60T and its helium volume held inside a larger envelope (Figure 2), the inertia and the water retained (Table S4). Some inputs come only from manufacturers, and the uncertainty of the historical airships is not estimated; since level 2 requires the uncertainty of all inputs to be at least estimated, input pedigree is level 1.

S5.2 Uncertainty characterization

The sources of uncertainty are those of Table S2. They are epistemic—the vertical inertia, the water a real unit retains, its response time, the hull volume of the LCA60T and the technology data—or aleatory—the outside-air temperature and the solar superheat of the lifting gas; the release size and D_{lim} are design choices, not uncertainties. The numerical uncertainty, below 0.002 %, is negligible against them. They are propagated by bounds, evaluating the model at the measured ends of each input, since the reported quantities are monotonic in each; with two inertia tests and six condensation experiments there is no distribution to sample. Propagating the measured spread of the heave mass over the whole vertical inertia gives an installed burn rate of 2.89–3.61 kg/s and a real-time rejection power of 407–509 MW; with the water collection measured by Zakhvatkin et al. [2] as well, at the four corners of both ranges, the rate is 3.25–4.34 kg/s and the hydrogen burned per tonne rises to 50.3–53.9 kg. Uncertainty characterization is level 2: most sources are quantified, classified and propagated by bounds; level 3 would require a probabilistic propagation of all of them.

S5.3 Results robustness

Equation (7) gives the local sensitivities of the installed rate in closed form, as elasticities $E_x = \partial \ln \dot{m}^*_{H_2} / \partial \ln x$. Taking logarithms, with $s = \sqrt{1 - m_{\mathrm{load}}/M_I}$, L and D_{lim} enter only outside s, and the payload and the inertia also through it, which gives

$$E_m = \frac{3}{2} + \frac{3(1-s)}{4s}, \qquad E_{M_I} = -\frac{1}{2} - \frac{3(1-s)}{4s}, \qquad E_L = -1, \qquad E_D = -\frac{1}{2}, \tag{S1}$$

with $E_m + E_{M_I} = 1$: scaling the payload and the inertia together leaves s unchanged. In the base case, a one-tonne release from the LCA60T, they are 1.50, −0.50, −1 and −1/2 (at most 1.52 and −0.52 on the Zeppelin NT). The heat-rejection power $P_{\mathrm{rej}} = \dot{m}^*_{H_2}\, q$ inherits them; the static requirements, hydrogen and heat per tonne, depend on neither the inertia nor D_{lim}. Through L, Equation (3), the water retained enters with $E_\phi = -r\phi/L = -0.40$ at $\phi = 1$.

Table S4 sets these against the range of each input. The release size is the strongest driver, and it is a design variable (Section S8). Of the uncertain inputs, the water a real unit retains moves the installed rate most, by +12–20 % with the collection measured by Zakhvatkin et al., then the inertia, by −9 to +14 %, and the hull volume of the LCA60T, by −5 %, or −17 % for a spheroid of its published length and diameter; none brings the heat-rejection power of a one-tonne release below hundreds of megawatts. The response time of the heat-rejection unit, $\tau = 5$ s, is of the order of the metal time constant estimated for aluminum air radiators of the FlyZero class (5.8–6.3 s [8]); a small laboratory exchanger with slow air, the only measured value, is slower (16–59 s [9]), but since the unit is started before the burn, a slower unit needs only an earlier start. Vertical drag is immaterial at the speeds of the maneuver: a drag coefficient in the range 0–1 changes the peak excursion by at most 0.5 %. The sensitivities to the other phenomena are in Table S2, with two combinations: the inertia with the water retained, at the corners of their measured ranges (Section S5.2), and the fan power with the specific power of the battery (Section S8). Results robustness is level 3: sensitivities are quantified for every phenomenon ranked high and for most of the others, with some combinations; level 4 would require many combinations.

Table S4: Sensitivity of the ideal bound of the base case (LCA60T, 1 tonne) to its inputs: nature of each input, range considered, elasticity of the installed rate $E_x = \partial \ln \dot{m}^*_{H_2} / \partial \ln x$ at the base case, Equation (S1), and installed rate, heat-rejection power and hydrogen per tonne over that range. The heat-rejection power is that of retaining $\phi = 0.99$, which needs the exhaust at 14 atm. Allowed sink: up to the largest value that meets the 1-m threshold. Hull volume: the other figure published for the LCA60T [26]; its elasticity is that of the inertia. Water retained: from the ideal bound to $\phi_{\max}$ at 1 atm times the collection measured by Zakhvatkin et al. [2]; elasticity at $\phi = 1$.

Input	**Nature**	**Range**	**Elasticity**	$\dot{m}^*_{H_2}$ **(kg/s)**	P_{rej} **(MW)**	H_2 **(kg/t)**
Release size m_{load}	design	1–20 t	1.50	3.17–288	446–40,613	44.8
Allowed sink D_{lim}	design	0.10–0.50 m	−0.50	1.42–3.17	199–446	44.8
Vertical inertia M_I	epistemic	0.77–1.20 of potential flow	−0.50	2.89–3.61	407–509	44.8
Hull volume, LCA60T	epistemic	180,000–200,000 m^3	−0.50	3.00–3.17	423–446	44.8
Water retained ϕ	epistemic	1 to 0.58–0.73	−0.40	3.17–3.81	446–503	44.8–53.9

S5.4 Technical review and process management of the use

The runs and their analysis, with their inputs, uncertainty and sensitivities, were reviewed informally by co-authors (level 1), and every run comes from the same pipeline and record as the development, under the same informally documented process (level 1).

S6 Outcome of the assessment and warnings

Table S5 and Figure S2 report the eleven factors against the threshold. The capability of the model reaches it in 3 of its 5 factors, and its results in 2 of 6. The gaps are of three kinds. The technical reviews and the process and product management concern how the work was examined and managed: a formal internal peer review and a formally documented and approved process would bring them to the threshold, and an independent review above it. The use assessment only further uses of the model can raise. The input pedigree concerns the data: the inputs that size the results carry their uncertainty, but those of the historical airships do not. None of the gaps changes a result. The first two leave open an error that the authors did not detect, which the automated checks that gate every number reduce and an independent review would address; estimating the uncertainty of the historical airships would give their results an uncertainty band, without changing the base case, the LCA60T.

The standard asks that results reported to decision makers carry their uncertainty, given in the article with each result, and explicit warnings wherever an assumption or a limit of the model is not met. For the results of the article the warnings are the six of its conclusions, given here with their detail:

1. The vertical dynamics of a full-size airship releasing a payload are not validated. The inertia rests on a potential-flow estimate checked against an 18-m^3 blimp, so the dynamic requirements—installed rate, real-time heat rejection and hull air flow—carry −9 to +14 %; the static ones do not.

2. The condensation bound and the black box of the heat-rejection unit favor the cycle (Table S1), so the power, response time and mass per kilowatt derived in the article are lower bounds of what real equipment must meet. The bound is validated against one fuel cell with excess air at 25–27 °C, not against a combustor at $\lambda = 1$ condensing at 40 °C; the cell retained 68–85 % of it, which would raise the hydrogen per tonne to 50.3–53.9 kg.

3. The installed rate carries no margin for the timing of the release, the estimation of the mass on board or gusts, which belong to the design of the controller.

4. The equipment masses are nominal, those of the lightest classes found, with no margin for growth. With the exhaust at 14 atm, a margin of 10 % on all of them costs 2.4 t of the 36 t that the LCA60T delivers in one-tonne releases of ten minutes, and the same margin on the heat to reject 2.2 t (Section 8).

5. In hold-down recoveries and deliveries in consecutive releases, lasting minutes to hours, the lifting gas is not isothermal: with the exhaust at 14 atm, its thermal drift, 0.6–1.2 t over the 10.6 min of one tonne with 10 MW and 3.0–6.0 t over the 6 h of a delivery on the LCA60T, is of the order of a release or of several and must be controlled on top of the releases and of the device (Section 9).

6. The results with a hold-down device (Section 5) assume ideal propellers, with a figure of merit of 0.7, and a device response bounded for a release from rest or for a force ramped from before it (Section 5.2). The response of a real device, and the static lightness it can be certified for (200 kg on the Zeppelin NT), limit the releases that can be held; container-scale releases can be held only by a moored airship.

Domain of validity The results are a sizing specification for the conceptual stage, built on inputs checked against reference data, a condensation bound validated against one experiment, a verified model and propagated input uncertainty; the integrated release of a full-size airship remains to be tested in flight.

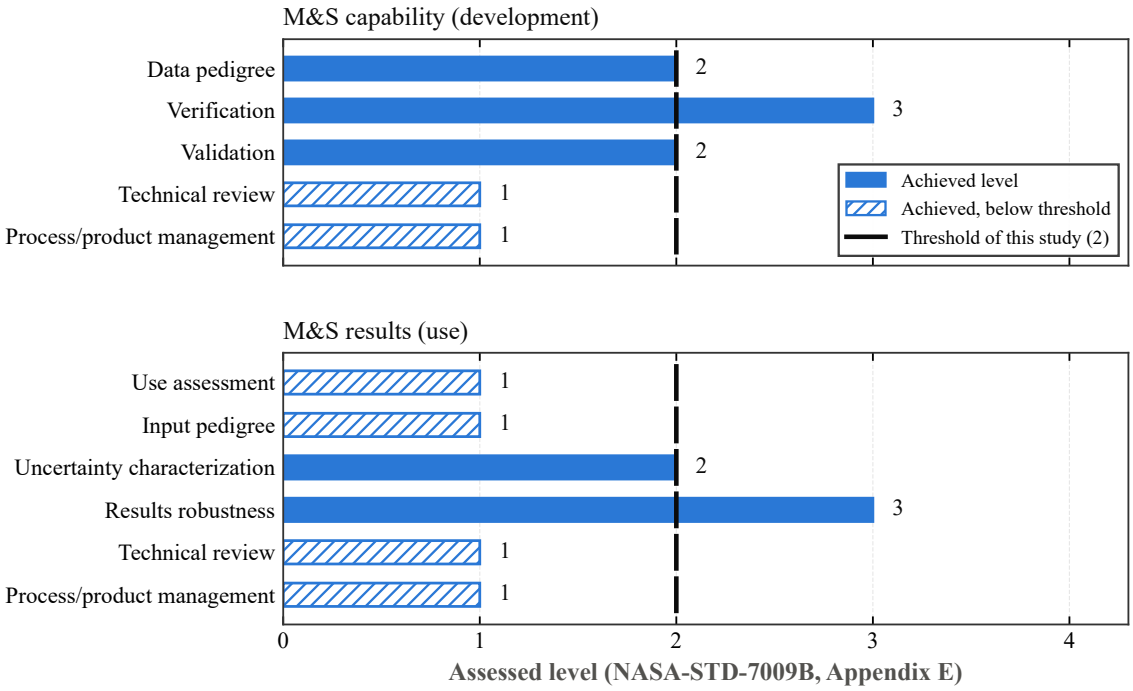


Figure S2: Credibility levels of the model, built in its development, and of its results, against the threshold of this study (NASA-STD-7009B, Appendix E). Hatched: below the threshold.

Table S5: Credibility assessment with the factors of NASA-STD-7009B [1]: level reached, from 0 to 4 (threshold of this study, 2), main evidence and what the next level would require. A level is a step without partial credit, reached only when all its conditions are met; the factors are not combined.

Factor	**Level**	**Evidence**	**Next level would require**
M&S capability (development)			
Data pedigree	2	All data traced to formal documents; 60 sources graded for pedigree (Table S3)	A referent representative of the hull for the added mass
Verification	3	Exact solution, convergence, closed form reproduced; 1122 automated checks gate the export	Reliable error estimation for the whole model, e.g. an independent implementation
Validation	2	Condensation bound against a representative system (Figure S1); inertia against blimp flight tests	Measurements on an airship releasing a payload
Technical review	1	Informal internal review by co-authors	Formal internal (2) or independent (3) peer review
Process/product management	1	One pipeline under version control, every number generated, export gated by the checks; process documented informally	A formally documented and approved process, roles and requirements
M&S results (use)			
Use assessment	1	Runs within the limits and the domain of verification of the model; first use of the model	Prior uses of the model
Input pedigree	1	Sizing inputs from formal documents, with estimated uncertainty (Table S4); historical airships without it	The uncertainty of all inputs, historical airships included
Uncertainty characterization	2	Sources classified, quantified and propagated by bounds (Section S5.2)	A probabilistic propagation of all sources
Results robustness	3	Closed-form elasticities; sensitivities of every phenomenon ranked high (Tables S4 and S2); two combinations	Many combinations
Technical review	1	Informal internal review by co-authors	Formal internal (2) or independent (3) peer review
Process/product management	1	The same pipeline and record for every run	As for the development

S7 Details of the ideal bound and of the altitude control

This section collects the details of the ideal bound (Section 4 of the article) and of its simulations that the article only summarizes.

Operation of the maneuver. Figure S3 shows the sequence for the base case. (1) The heat-rejection unit is started 22 s before the release, 3 response times before the preconditioning burn, which bring it to 95 % of its capacity (Section 3.4), so that its response time does not limit the burn. (2) 7.1 s before the release, the anticipation time of the closed-form sizing (Section 4), the conversion unit burns hydrogen at the installed rate: the airship sinks by at most D_{lim} and pre-compensates a fraction α of the payload, about half. (3) At release the altitude controller takes over and keeps burning at the installed rate for about 7 s, until the gas mass reaches its new equilibrium and the airship comes to rest, slightly below the release point. (4) The controller then trims the gas volume at a low rate while the airship returns to its reference altitude. The two phases are one burn at the installed capacity, sized so that nothing is burned beyond the new equilibrium.

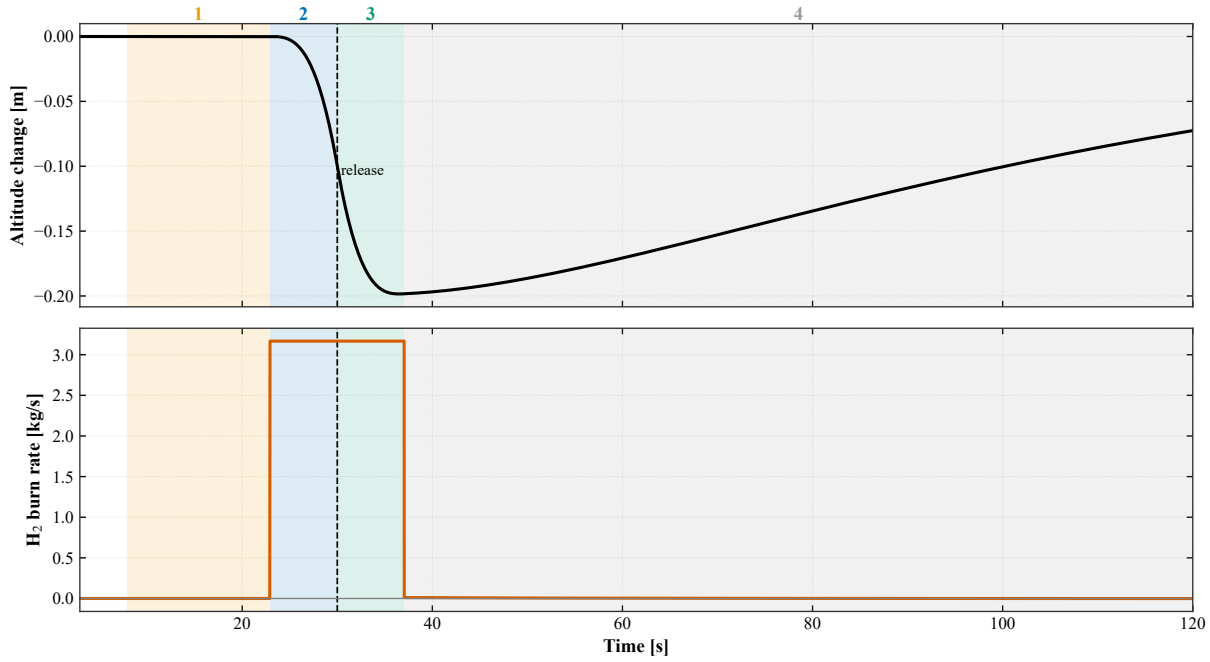


Figure S3: Operation of the maneuver, base case (LCA60T, 1 tonne, 1-m threshold, instantaneous condensation): (1) heat-rejection unit started, (2) preconditioning burn, (3) post-release burn under the altitude controller, (4) trim. Dashed line: release. With instantaneous condensation step (1) has no effect on the maneuver; it is shown to set the sequence used with a real unit in Section 5.

Reactive burn without preconditioning. The preconditioning is what keeps the installed rate as low as $\dot{m}^{*}_{H_2}$. A burn that starts only at the release, at the full rate from that instant, lets the vehicle rise $h = 2\,g\,m_{\mathrm{load}}^{3}/[3\,(M_I - m_{\mathrm{load}})\,L^2\,\dot{m}^2]$ before stopping it, so keeping the rise below h_{max} needs $\sqrt{4\,(1+s)^3 D_{\mathrm{lim}}/(s^2 h_{\mathrm{max}})}$ times $\dot{m}^{*}_{H_2}$: 1.8 times for the 1-m threshold and 4 times to stay within the $(1+s)\,D_{\mathrm{lim}}$ of the preconditioned burn. The altitude controller of the article (Appendix B), acting only after the release, needs 2.8–3.0 and 12–13 times in the simulations (Table S6), 9.0 and 40 kg/s in the base case. Only the 5-m safety threshold admits a reactive burn below $\dot{m}^{*}_{H_2}$, 0.84–0.88 times, at the cost of a 5-m rise right after the release. The thresholds therefore size a reactive burn, whereas the preconditioned burn meets them by itself and only D_{lim} sizes it. Figure S4 shows both burns over the whole range of installed rates: for the same excursion, the reactive burn needs 4.2–4.4 times the

rate of the preconditioned burn at 5 m, 6.3–6.7 times at 1 m and 12–13 times at the $(1+s)\,D_{\mathrm{lim}}$ of the preconditioned burn.

Table S6: Installed rate that the altitude controller needs without preconditioning, with the same water retention, to keep the rise within 1 m or within the $|\Delta h|_{\mathrm{peak}}$ of the preconditioned burn, in kg/s and as a multiple of the closed-form rate of Equation (7) (Table 3).

	Installed rate (kg/s)		**Times $\dot{m}^*_{H_2}$**	
Airship	**1-m rise**	**Same $\|\Delta h\|$**	**1-m rise**	**Same $\|\Delta h\|$**
LZ 129 Hindenburg	8.0	35	2.8	12
Flying Whales LCA60T	9.0	40	2.8	13
ZR-3 Los Angeles	13.7	61	2.9	13
Zeppelin NT	44.1	198	3.0	13

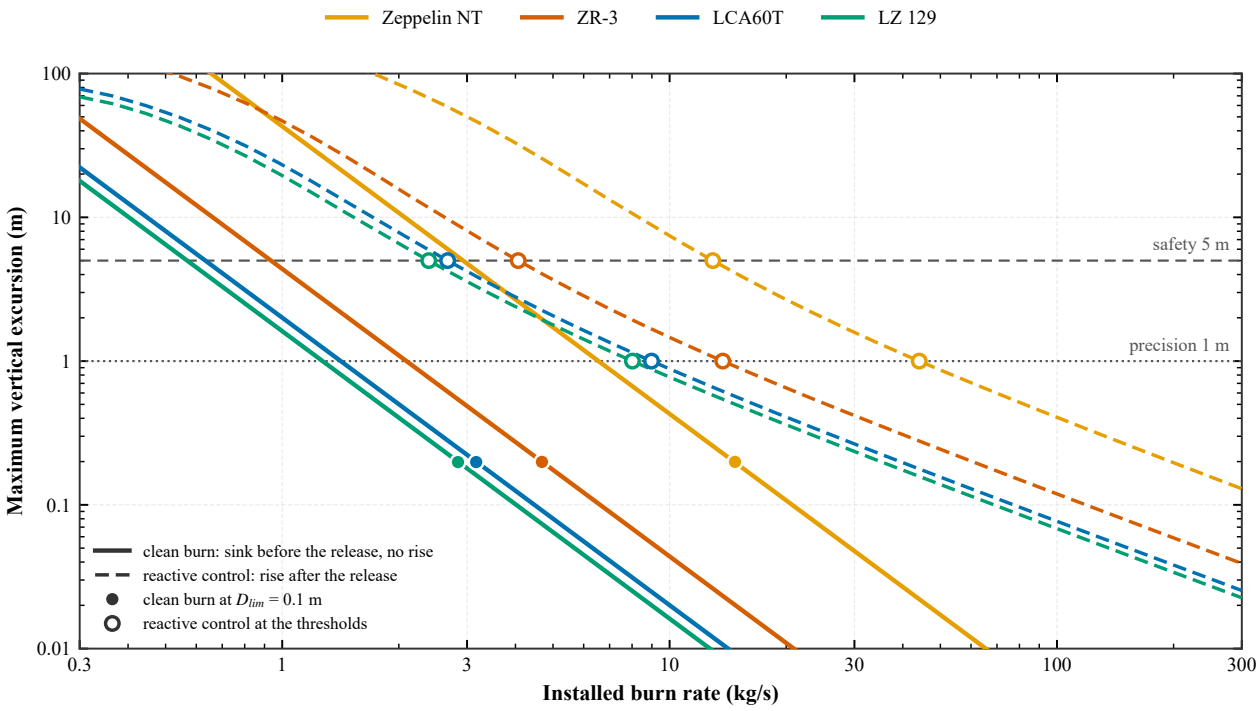


Figure S4: Maximum vertical excursion versus installed burn rate, 1-tonne release, instantaneous condensation. Solid: preconditioned burn, which sinks $(1+s)\,D_{\mathrm{lim}}$ before the release and does not rise (Equation (7)); dashed: reactive burn, which rises after the release (altitude controller of Appendix B, simulated). Filled dots: preconditioned burn at $D_{\mathrm{lim}} = 0.10$ m; open dots: reactive burn at the 1-m and 5-m thresholds (Table S6).

Choice of D_{lim}. The rate scales as $D_{\mathrm{lim}}^{-1/2}$: the largest value that still meets the 1-m threshold, $h_{\max}/(1+s) = 0.50$ m, gives 1.42 kg/s and 199 MW of heat rejection in the base case, but the vehicle then reaches the release sinking at 0.095 m/s instead of 0.042 m/s.

Burning past the equilibrium. A lower rate requires burning past the equilibrium and returning the excess hydrogen from the tanks. Burning at the full rate until the vehicle, rising after the release,

peaks at $h_{\max} = 1$ m above h_0 lowers the installed rate by 38 % in the base case, to 1.98 kg/s, but burns 52 % more hydrogen than the equilibrium, which must then be returned to the gas cells.

Stiffness of the altitude loop in the simulations. In the simulations the lift follows the air density at the current altitude with the gas volume fixed, which adds 61 % to the stiffness of the altitude loop in the base case. With the gas cells about half full, the gas would instead expand or contract with the outside pressure and keep its lift in either direction, so this stiffness is an artefact of the model: it makes the trim faster than in the airship; the excursions reported, set by the saturated burn, are not affected.

S8 Mass budget, heat-rejection classes and power versus release size

This section gives the detail behind Section 5.1 of the article: the mass budget of the base case at 1 atm, without the exhaust compressor, as in that section, with the representative class of each black box (Table S7), the heat-rejection classes found, against the requirement at 1 atm (Table S8), and the power needed to hold the ideal bound for releases larger than the reference tonne (Table S9). The article summarizes the budget in Figure 5, with the lightest class of each box, for the four airships and with a hold-down device.

Table S7: Mass budget per kg/s of hydrogen burned for the ideal bound at 1 atm, where the exhaust retains 0.85 of its water and no exhaust compressor is carried (LCA60T, 1-tonne release). The budget isolates the payload available for the unselected heat-rejection unit (Table S8) by subtracting the tonne to be released, the fixed masses (assuming a 24-h regeneration cycle) and the scaling components (a burner with a battery driving its fans). Ranges run from the lightest to the heaviest class of each box.

Black box	**Representative class**	**Mass**	**Source**
Scale with the burn rate (t per kg/s of hydrogen)			
Energy conversion unit	H_2 reciprocating engine, $\lambda = 1$	26–126	[27, 28]
Energy buffer	Li-ion pack (power-sized)	16–42	[29]
Heat-rejection unit	Not chosen (budget remainder)	≤ 1.5	–
Fixed, per tonne released (t)			
Regeneration unit	Aviation PEM stack	0.04–0.18	[30]
H_2 tanks	Type IV vessels	0.87–0.91	[31]
H_2 compressor	Electric industrial unit	0.55	[32]
Water tank	Collapsible fabric tank	0.001–0.003	[33]
Budget for the boxes that scale: (60 t − 1 t − 1.45 t) / 3.37 kg/s		17.1	

Real-time condensation, in numbers. At the 0.653 kW/kg of air-cooled condenser bundles, the 100-MW case of Figure 3 of the article would weigh 153 t of bundles. In the simulations of that figure, at 1 atm, the heat-rejection unit of the LCA60T starts 22 s before the release, with $\tau = 5$ s, and with 10 MW the airship rises 2.50 m. In the ideal bound the 45 kg of hydrogen burned on the LCA60T leave 401 kg of water on board; with 10 MW at 1 atm only 24 kg are retained, 795 kg leave with the nitrogen, and the controller burns 92 kg of hydrogen.

The single release with a hold-down device, in numbers. With the exhaust at 14 atm, on the three larger airships the largest single release reaches 16–17 % of the payload with a one-hour recovery (Figure 4 of the article, divided by the payload). Without the exhaust compressor, at 1 atm,

Table S8: Heat-rejection classes that exist, against the requirement of Section 5.1 of the article (base case at 1 atm, burner): heat rejected per unit mass that each class has, taken as the whole unit except for the finned-tube bundles, given alone, and that the cycle needs at the fan power f of that class, from a CT7-class turbine to the lightest battery class driving the fans (with fans sized for the pressure drop, $f = 0.008$–0.041, the requirement of the article is 8–11 kW/kg); gap: needs over has, from the most to the least favorable (below 1 the class fits). Classes grouped by the temperature at which they reject the heat: above the condensation band of the cycle (40–73 °C at 1 atm) or within it. TMS: thermal-management system; ACC: air-cooled condenser; T: temperature at which the heat is rejected; f: fan (or drag) power as a fraction of the heat rejected. Rigid airships: equipment mass per kg/s of water recovered at 50 kn, converted with the 18.1 MJ per kg of water of this cycle at 1 atm; in parentheses, at full power and with all the water of methane. [a]Not reported; taken as 0, the most favorable. [b]The battery that drives the fans alone exceeds the budget, so no mass per kilowatt is enough: the class needs no finite value ("none") and its gap is unbounded (∞).

Candidate	T **(°C)**	**Air**	f **(%)**	**Has** (kW/kg)	**Needs**	**Gap** (×)	**Source**
Rejecting above the condensation band							
Aircraft TMS, 2026	100	ram	20	5.0	10–none[b]	2.0–∞	[8]
Aircraft TMS, 2035	100	ram	15	10	9.6–93	0.96–9.3	[8]
Aircraft air radiator only, 2026	80	ram	20	10	10–none[b]	1.02–∞	[8]
Aircraft air radiator only, 2035	160	ram	15	20	9.6–93	0.5–4.7	[8]
Condensing at the temperature of the cycle							
Power-plant ACC, whole system	57	fans	1.0	0.135	8.2–8.6	60–64	[34]
Power-plant ACC, finned-tube bundles	57	fans	1.0	0.653	8.2–8.6	12–13	[34]
Rigid airships, water recovery	exhaust	ram	–[a]	0.304–0.566 ($\leq$ 2.8)	8.1	2.9–27	[4]

Table S9: Heat-rejection power (MW) needed to hold the ideal bound for single releases of 1 to 20 tonnes ($D_{\mathrm{lim}} = 0.10$ m, retaining $\phi = 0.99$, which needs the exhaust at 14 atm), up to the payload capacity of each airship.

	Release (t)				
Airship	**1**	**2**	**5**	**10**	**20**
LZ 129 Hindenburg	400	1133	4490	12,746	–
Flying Whales LCA60T	446	1263	5006	14,224	40,613
ZR-3 Los Angeles	659	1869	7432	–	–
Zeppelin NT	2077	–	–	–	–

the boxes that scale with the heat rejection weigh 1.7 t per MW and the heat per tonne is 6.57 GJ. At 14 atm, on the LCA60T a ten-tonne release takes at least 60 min to recover and a twenty-tonne release at least 5.0 h.

Sharing the release with a hold-down device. Table S10 details, on the LCA60T, how the release is shared between the burn and the device for each installed heat rejection and releases of 1, 10 and 30 tonnes.

Table S10: Sharing the release with a hold-down device on the LCA60T, with the exhaust at 14 atm (Section 5.2 of the article). For each installed heat rejection P_{cap}: the part of the release absorbed by the burn alone (m_e); the recovery time (Equation (8)), which does not depend on the airship; and the force the device must take with its peak power as ideal propellers (disk area 20–100 m^2, figure of merit 0.7), which a tether replaces with no power. The table stops at 39 MW, where the condenser bundles weigh the whole payload.

				Hold-down device	
Release	P_{cap} **(MW)**	**Absorbed by the burn,** m_e **(kg)**	**Recovery time**	**Force (kN)**	**Power (MW)**
1 t	2	27	52.8 min	9.5	0.09–0.19
	10	80	10.6 min	9.0	0.08–0.18
	39	197	2.7 min	7.9	0.06–0.14
10 t	2	27	8.8 h	97.8	2.8–6.2
	10	79	105.7 min	97.3	2.8–6.2
	39	196	27.0 min	96.1	2.7–6.1
30 t	2	27	26.4 h	293.9	14.5–32.5
	10	78	5.3 h	293.4	14.5–32.4
	39	194	81.0 min	292.3	14.4–32.3

Each release against its recovery time. Figure S5 gives Equation (11) of the article on each airship, with the exhaust at 14 atm, against the recovery time of each release rather than the time of the whole delivery. In one-tonne releases the LCA60T delivers 36 t of delivered cargo with ten minutes per release and 51 t with thirty, up to 58.5 t. The LZ 129 delivers 10 t in one-tonne releases of thirty minutes and the ZR-3 3.3 t in releases of a quarter tonne recovered in an hour. The Zeppelin NT delivers 0.34 t in releases of 25 kg recovered in an hour and at most 0.43 t, held down by its catalog compressor; one that shrank with the release would leave it 0.87 t. The shading marks which hold-down device can take each release: a tether everywhere, and propellers powered by the cycle's engine alone in the white region.

Releases as a fraction of the payload. With the exhaust at 14 atm and every box proportional to the release size, the hydrogen compressor in proportion to its capacity ($b_{\mathrm{lin}} = b' + m_{\mathrm{c}}$ per 1.6 t = 1.25 t per tonne), Equation (11) divided by the payload becomes $N/C = [1 - f\,(b_{\mathrm{lin}} + a\,Q/t_{\mathrm{rec}})]/(1 + b_{\mathrm{w}})$, with $f = m_{\mathrm{load}}/C$ the release size as a fraction of the payload and b_{w} the water tank per tonne delivered, the same on every airship (Figure S6). It is the limit of the large airships, whose larger releases need several compressor units. Each curve starts on the single release of Equation (10), where one release is all the cargo, and rises as the recovery lengthens: with releases of 1 % of the payload recovered in thirty minutes, 91 % of it is delivered cargo. Below 1.6 t, the hydrogen compressor is a single catalog

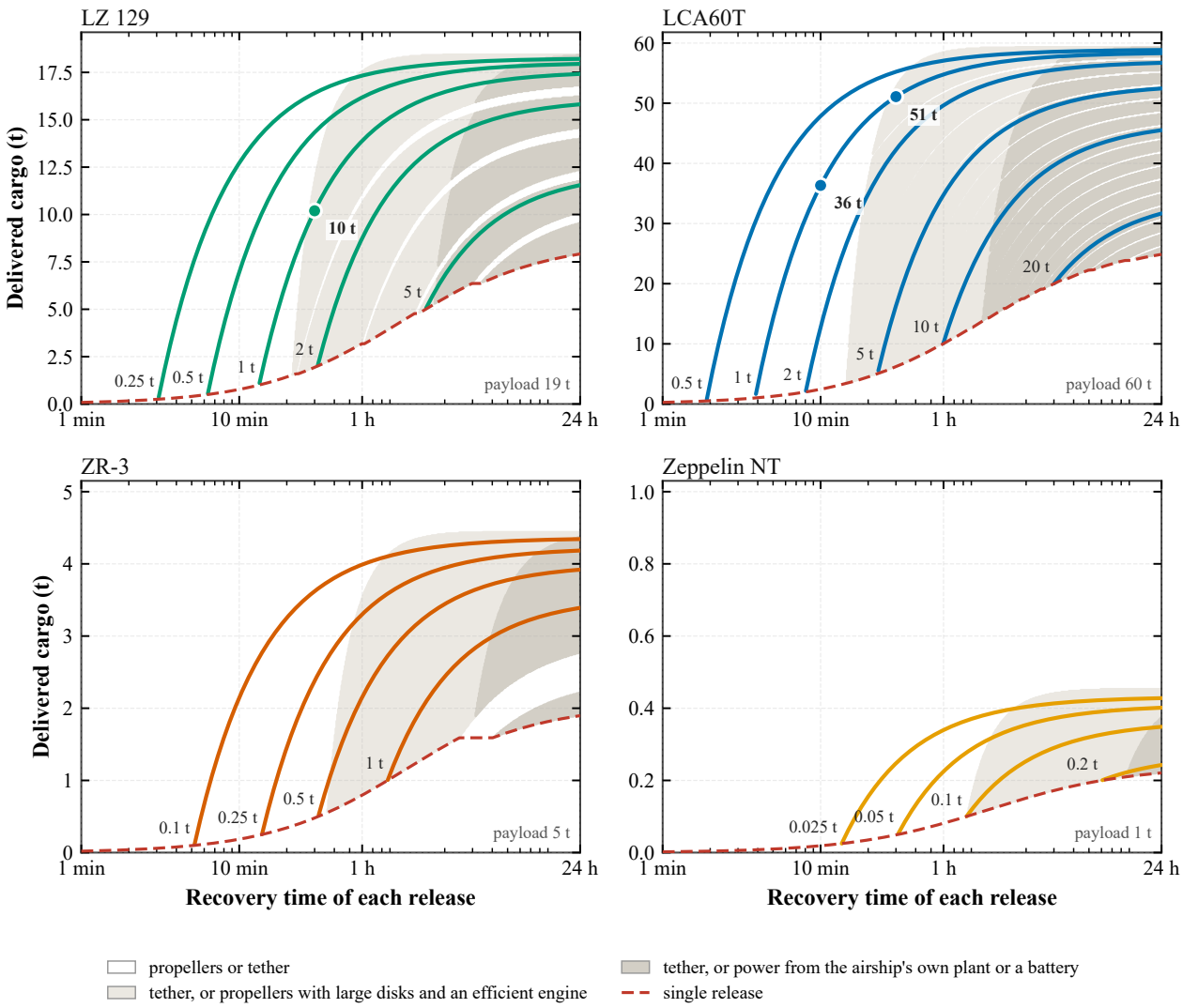


Figure S5: Delivered cargo delivered in releases of the size marked on each curve, with the same equipment and the exhaust at 14 atm, versus the recovery time of each release (Equation (11) of the article). Shading: hold-down device that can take each release, a tether everywhere and propellers powered by the cycle's engine alone in the white region; red: single release; dots: one-tonne releases recovered in ten and thirty minutes.

unit that does not shrink with the release size, and it lowers the curves of each airship by its share of the payload.

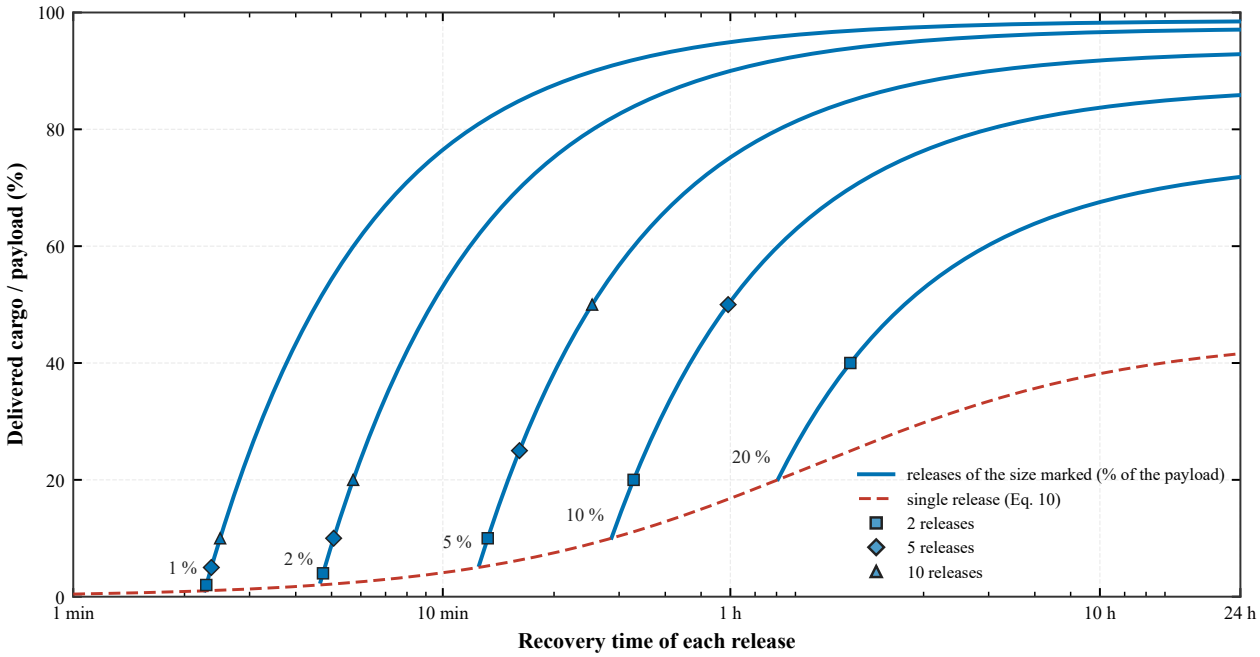


Figure S6: Delivered cargo, as a fraction of the payload, delivered in releases of the fraction of the payload marked on each curve versus the recovery time of each release, with every box proportional to the release size, the hydrogen compressor in proportion to its capacity and the exhaust at 14 atm. Red: single release; markers: two, five and ten releases.

S9 Energy of the release and of the regeneration

This section gives the numbers behind Section 6 of the article. With 10 MW of installed heat rejection, the work left for the plant after the fans, the propellers and, at 14 atm, the exhaust compressor is 194–700 kWh per tonne at 1 atm and 77–578 kWh at 14 atm, where the exhaust compressor takes 82 kWh. Regenerating a tonne released at 1 atm takes 2147 kWh for the electrolysis (53 kWh/kg [10]) and 82–126 kWh to compress the hydrogen into the tanks (2.0–3.1 kWh/kg [11, 12]). A regeneration module on hull panels allowed one week of sun costs 1.37–1.52 t of payload per tonne recovered at 14 atm with amorphous silicon and 2.95–3.11 t with CIGS.

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